EXISTENCE OF THE SURJECTIVE MAP ONTO THE KERNEL OF THE FIRST COMPARISON MAP BETWEEN THE MODULES ATTACHED TO CONSECUTIVE PARAMETERS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned article source sources/sigma-2014-110-demazure-fusion/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The cyclic graded modules V_{i,k} with their five families of defining relations, labelled r1 to r5 in the article, and the shift operation tau. The author's section 4.2, where the modules and their relations are set up before the counted statement; the five relations are referred to in the counted proof (source lines 641-648) as navigation-pointer stubs. Supplies the relations against which the map has to be checked, and the module on which it is defined by the value on one generator.
- The two relations of the local Weyl module and of the Demazure module recorded in the lemmas labelled g and crucial. Both are invoked at the end of the counted proof (source line 649) as navigation-pointer stubs, each being counted separately in this lane. Identify the last two defining relations with statements already proved for the current algebra modules.

Source reference. Original reference r1: sources/sigma-2014-110-demazure-fusion/pinned-original-source.tex, source lines 641–641. Navigation pointer only: the counted text refers to the item r1 of this same article, which is not part of this card. The pointer carries the item's own author label as its number; its content is not reproduced here.

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Source reference. Original reference g: sources/sigma-2014-110-demazure-fusion/pinned-original-source.tex, source lines 492–492. Navigation pointer only: the counted text refers to the item g of this same article, which is not part of this card. The pointer carries the item's own author label as its number; its content is not reproduced here.

Source reference. Original reference crucial: sources/sigma-2014-110-demazure-fusion/pinned-original-source.tex, source lines 649–649. Navigation pointer only: the counted text refers to the item crucial of this same article, which is not part of this card. The pointer carries the item's own author label as its number; its content is not reproduced here.

Proposition 1. *There exist a surjective morphism of* $\mathfrak{g}[t]$*-modules* $\phi^- : \tau_{2k+1-i}\mathbf{V}_{i,k} \to \ker \phi^+$*, such that* $\phi^-(\overline{v}_{i,k}) = \big(x_\theta^- \otimes t^{2k+1-i}\big)\overline{v}_{i,k+1}$.

Proof. We only need to show that $\phi^-(\overline{v}_{i,k})$ satisfies the defining relations of $\mathbf{V}_{i,k}$. We start with the relation (r1). First, for $\alpha = \theta$ it is clear. Let $\alpha \in R^+ \setminus \{\theta\}$; if $(\theta|\alpha) = 0$ then also it is clear. If $(\theta|\alpha) = 1$ then $(\theta - \alpha) \in R^+ \setminus \{\theta\}$ and $(\theta|\theta - \alpha) = 1$, now it is clear from the relations $(x^-_{\theta-\alpha} \otimes t^r)\overline{v}_{i,k+1} = 0$ for all $r \geq (k+1)$ in $\mathbf{V}_{i,k+1}$. The relations (r2), (r3) are trivially satisfied by $\phi^-(\overline{v}_{i,k})$. Finally the last two relations (r4), (r5) are also satisfied by $\phi^-(\overline{v}_{i,k})$; in fact these are exactly the statements of Lemmas g and crucial. □